Trigonometry to Find Lengths

Exam code: 0607

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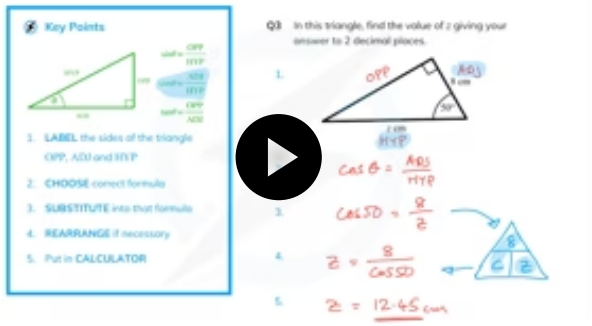

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Help me understand

SOHCAHTOA – Finding lengths

What is trigonometry?

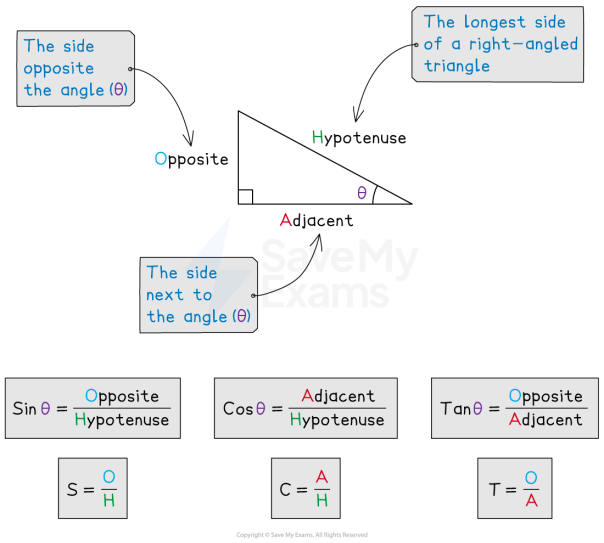
- Trigonometry is the mathematics of **angles** in triangles
- It looks at the relationship between **side lengths** and **angles of triangles**
- It comes from the Greek words *trigonon* meaning 'triangle' and *metron* meaning 'measure'

What are sin, cos and tan?

- The **three** trigonometric functions **sine**, **cosine** and **tangent** come from **ratios of side lengths** in **right-angled** triangles
- To see how the ratios work you must first **label the sides** of a **right-angled triangle** in relation to a **chosen angle** θ
  - The **hypotenuse, H,** is the **longest side** in a right-angled triangle
    - It will always be **opposite** the right angle
  - The side **opposite θ** will be labelled **opposite**, **O**
  - The side **next to θ** will be labelled **adjacent**, **A**
- The functions sine, cosine and tangent are the ratios of the lengths of these sides as follows
  - $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{\text{O}}{\text{H}}$

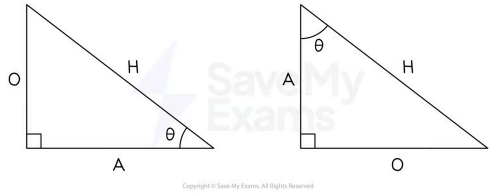
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

adjacent A



What is SOHCAHTOA?

- **SOHCAHTOA** is a mnemonic that is often used as a way of remembering which ratio is which
  - **S**in is **O**pposite over **H**ypotenuse
  - **C**os is **A**djacent over **H**ypotenuse
  - **T**an is **O**pposite over **A**djacent
- **H** is always the same but **O** and **A** change depending on which angle is labelled as θ



How can I use SOHCAHTOA to find missing lengths?

- **STEP 1**
  Label the sides of the triangle as H, O and A
  - **H** is the **longest side** opposite the right angle
  - **O** is **opposite** the given angle
  - **A** is **next to** the given angle
- **STEP 2**
  Identify which t**rigonometric ratio** to use: sin, cos or tan
  - Write down the letter of the **length** you are **given**
  - Write down the letter of the **length** you **want to find**
  - Find the two letters in **SOHCAHTOA** to identify which ratio to use
    - If you have **A** and **H** then use **cos**
- **STEP 3**
  Substitute the values into the relevant trigonometric formula

- STEP 4
  Rearrange and **solve** for the **unknown letter**
  - You will either need to **multiply** or **divide**
    - $\sin(50) = \frac{A}{7}$ leads to $A = 7 \times \sin(50)$
    - $\cos(40) = \frac{3}{H}$ leads to $H = \frac{3}{\cos(40)}$
- STEP 5
  Type the expression **into your calculator**
  - The **question** might ask you to **round your answer**
  - If not then round to **three significant figures**

Examiner Tips and Tricks

- SOHCAHTOA (like Pythagoras) can only be used in **right-angled triangles**
- Ensure your calculator is set to measure angles in **degrees**
  - You should see the letter **D** or the word **Deg** at the top of your screen

Worked Example

Find the length of the side x cm in the following triangle.

Give your answer to 3 significant figures.

Answer:

First label the triangle

We know A and we want to know O - that's TOA or

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

Multiply both sides by 9

$$9 \times \tan(43) = x$$

Enter on your calculator

$$x = 8.3926...$$

Round to 3 significant figures

$$x = 8.39 \text{ cm}$$

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